House Price Prediction Using Machine Learning

Abstract

This project develops a machine learning model to predict house prices using selected housing features from a dataset of 1,460 houses.

Objectives

To analyse housing data, train a regression model, predict house prices, and evaluate prediction performance.

Dataset

The dataset contains 1,460 rows and 82 columns. Five numerical features were selected to predict SalePrice.

Features Used

OverallQual, GrLivArea, GarageCars, TotalBsmtSF and YearBuilt.

Methodology

The data was divided into 80% training data and 20% testing data. A median imputer and Linear Regression model were used in a pipeline. The trained model was evaluated on the testing data.

Results

Mean Absolute Error (MAE): 25,414.73
Root Mean Squared Error (RMSE): 39,763.30
R-squared Score: 0.7939
Sample predicted house price: 234,913.95

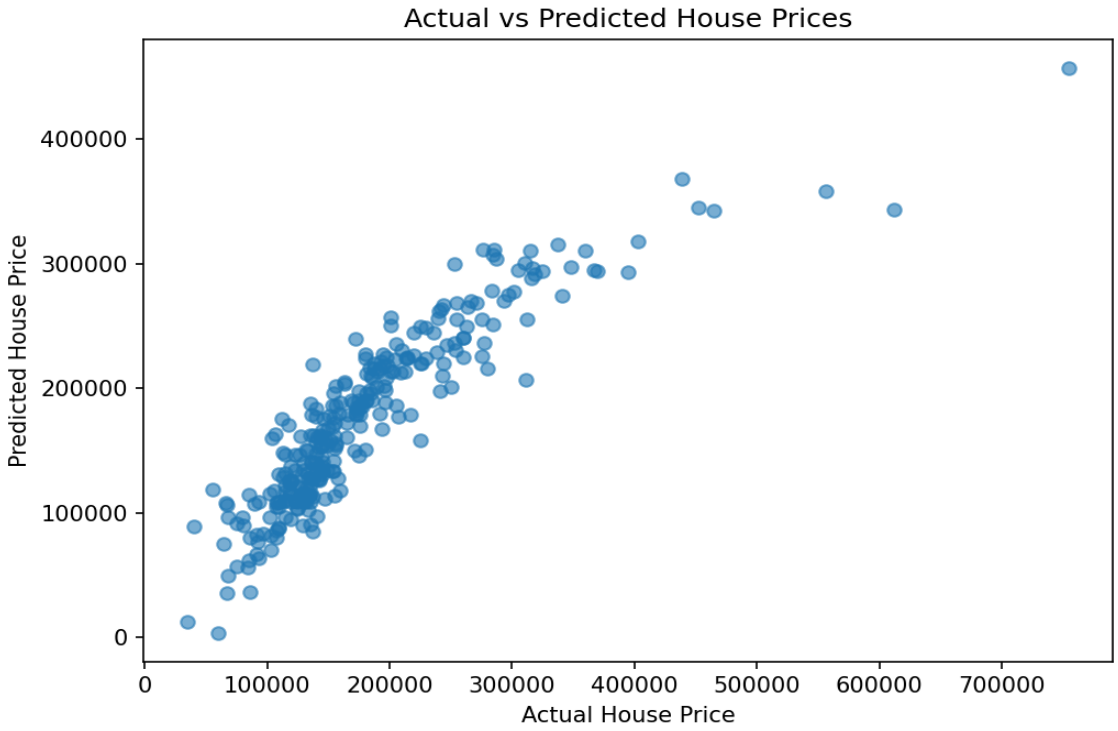
Conclusion

The Linear Regression model achieved an R-squared score of 0.7939 on the test set. The model demonstrates useful predictive ability, although individual predictions may differ from actual house prices.

Limitations

The model uses only five features. Adding and validating more relevant features and comparing other regression models may improve performance.

Actual vs Predicted House Prices


Actual vs Predicted House Prices
Predicted House Price
Actual House Price
0
100000
200000
300000
400000
500000
600000
700000